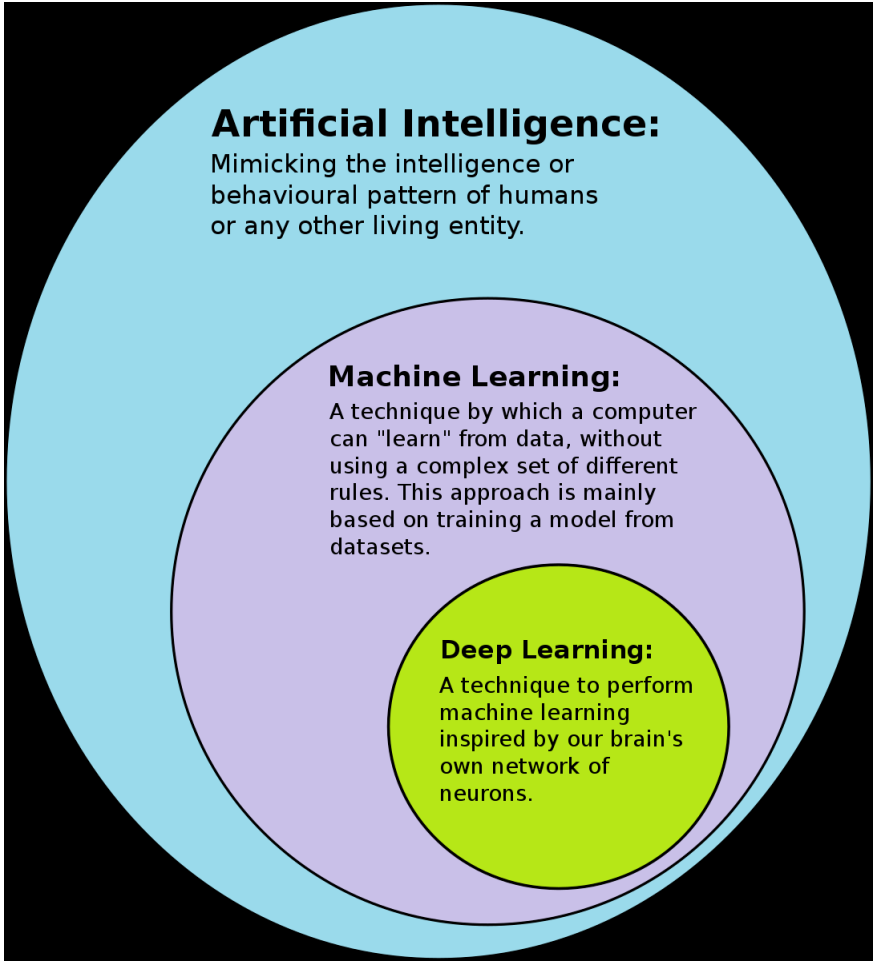
Software Technology 06/B

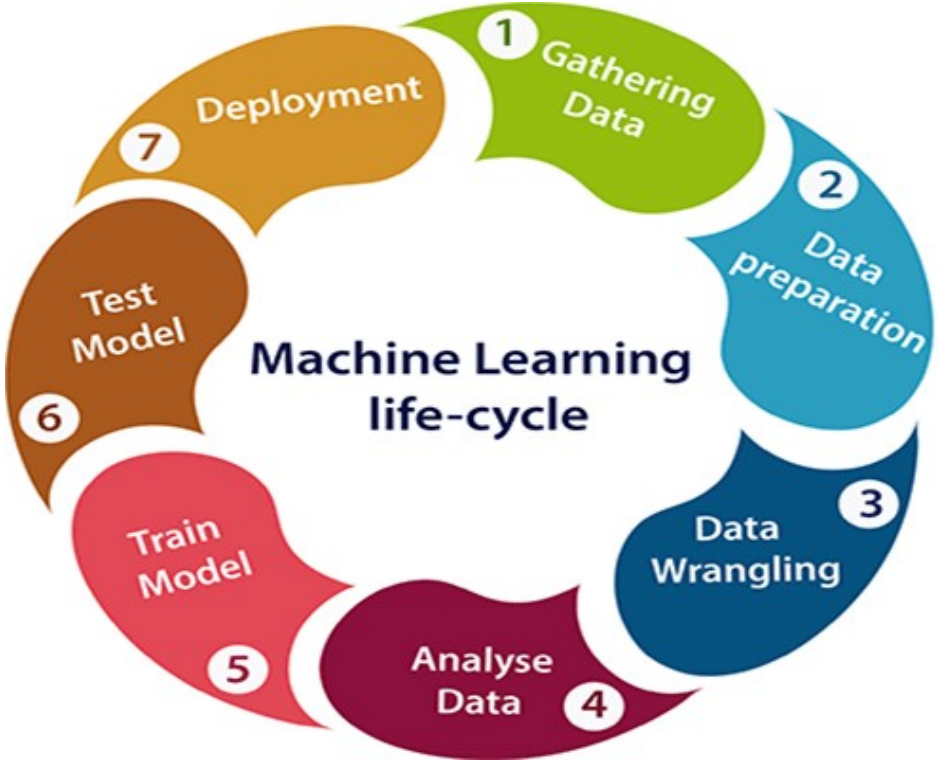
MLOps

AI, ML, ...


Artificial Intelligence:
Mimicking the intelligence or behavioural pattern of humans or any other living entity.
Machine Learning:
A technique by which a computer can "learn" from data, without using a complex set of different rules. This approach is mainly based on training a model from datasets.
Deep Learning:
A technique to perform machine learning inspired by our brain's own network of neurons.

Programming vs. ML

ML life-cycle

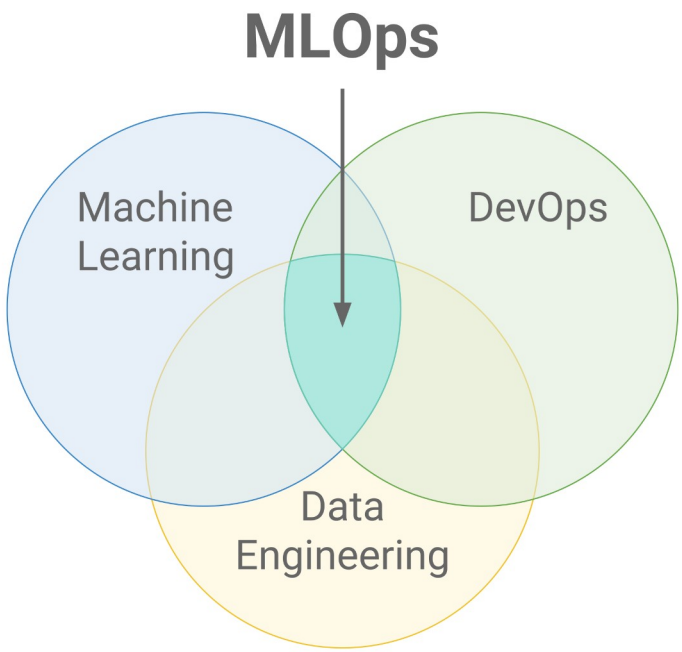


Why MLOps?

- High hopes:
  - Everyone wants ML
    - Partly a hype
    - Partly high business potential
- “A large number of ML projects fail – with many ML proofs of concept never progressing as far as production” (source)
  - Access to data (quality, quantity)
  - Maintenance is hard and costly
    - External world changes → “Model drift”
    - Retraining

MLOps

- Deploy & maintain ML models
- Traditional SW: DevOps
- ML models: MLOps
- Increase automation
- Improve quality
- ML life-cycle management
  - Model creation
  - CI/CD, orchestration
  - Monitoring: health, diagnostics…



- Note: MLOps ≠ AIOps
  AIOps (AI for IT operations) is the application of AI to automate and streamline operational workflows (e.g., log analysis, ticket sorting, RCA)

Bridging the gap

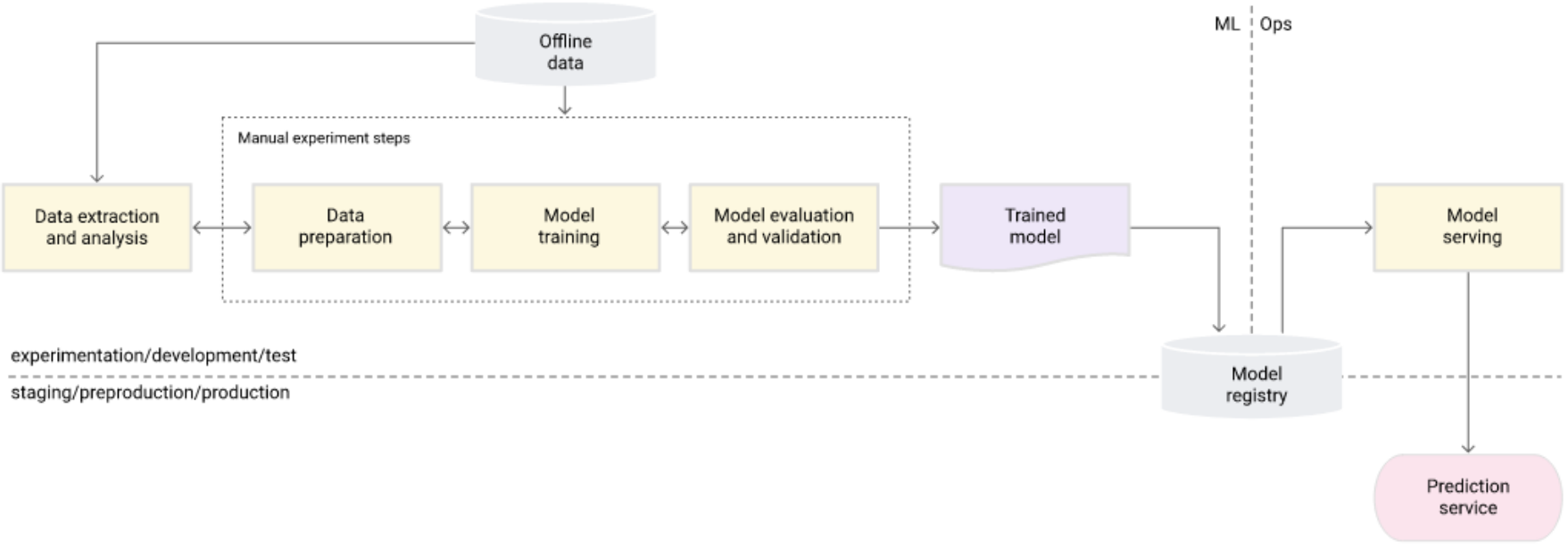
- Data scientists
- PoC environment
- Code in notebooks (Jupyter etc.)

- Operations team
- Production environment
- Code in microservices (Non-functional reqs)

MLOps maturity levels

- Levels of automation…
- Different proposals:
  - Google:
    “MLOps: Continuous delivery and automation pipelines in machine learning”
    (The following slides will present this one…)
  - Microsoft:
    ”Machine Learning operations maturity model”

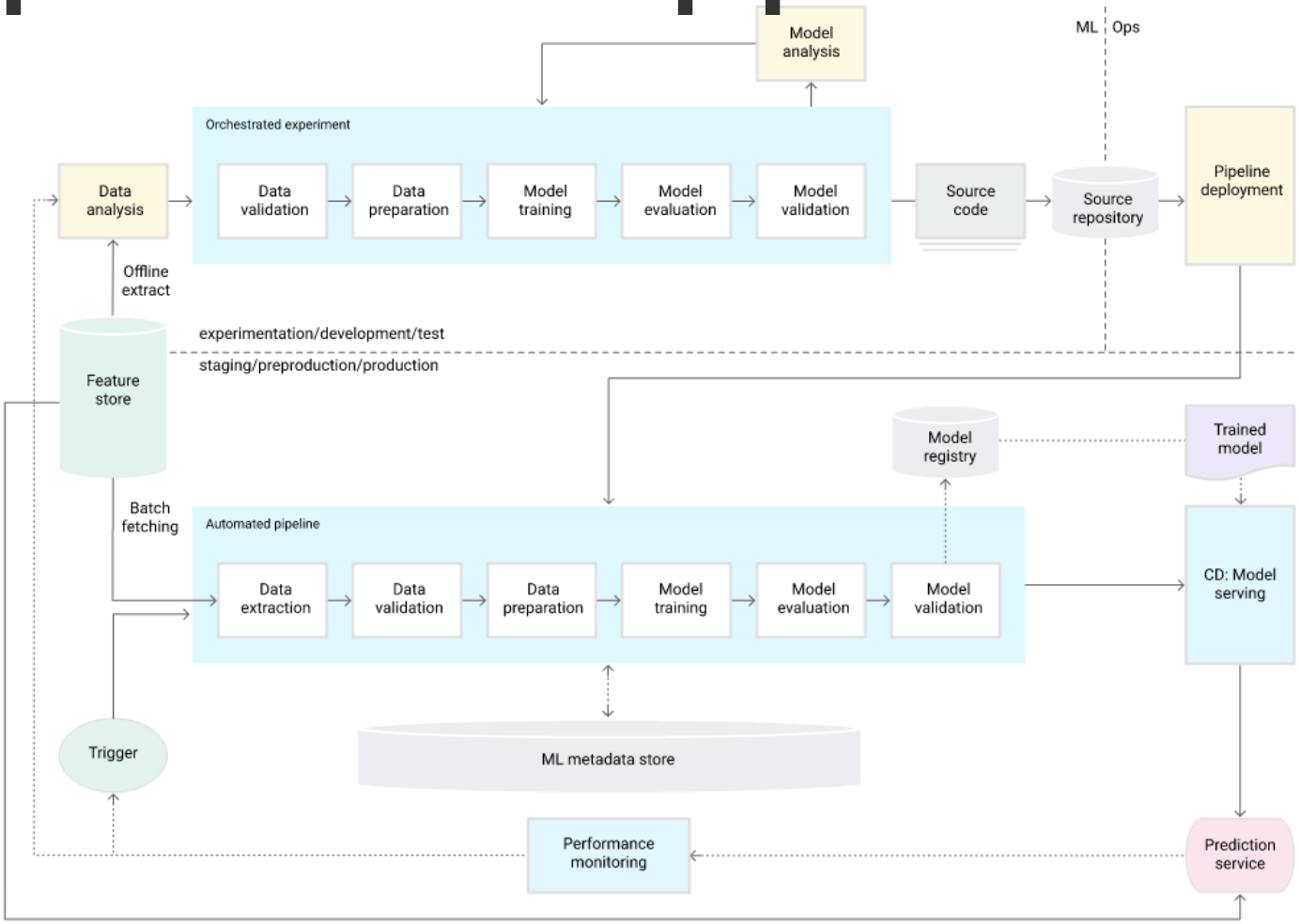
MLOps level 0: Manual process



MLOps level 0: Manual process

- Every step is manual, driven by experimental code that is written and executed in notebooks by data scientist
- Separates data scientists who create the model and engineers who serve the model as a prediction service
- Infrequent release iterations
- No CI&CD
- Concerned only with deploying the trained model rather than the entire ML system
- Lack of active performance monitoring
- → **For businesses beginning to apply ML. For models that rarely change. Models are expected to break in production.**

MLOps level 1: ML pipeline automation

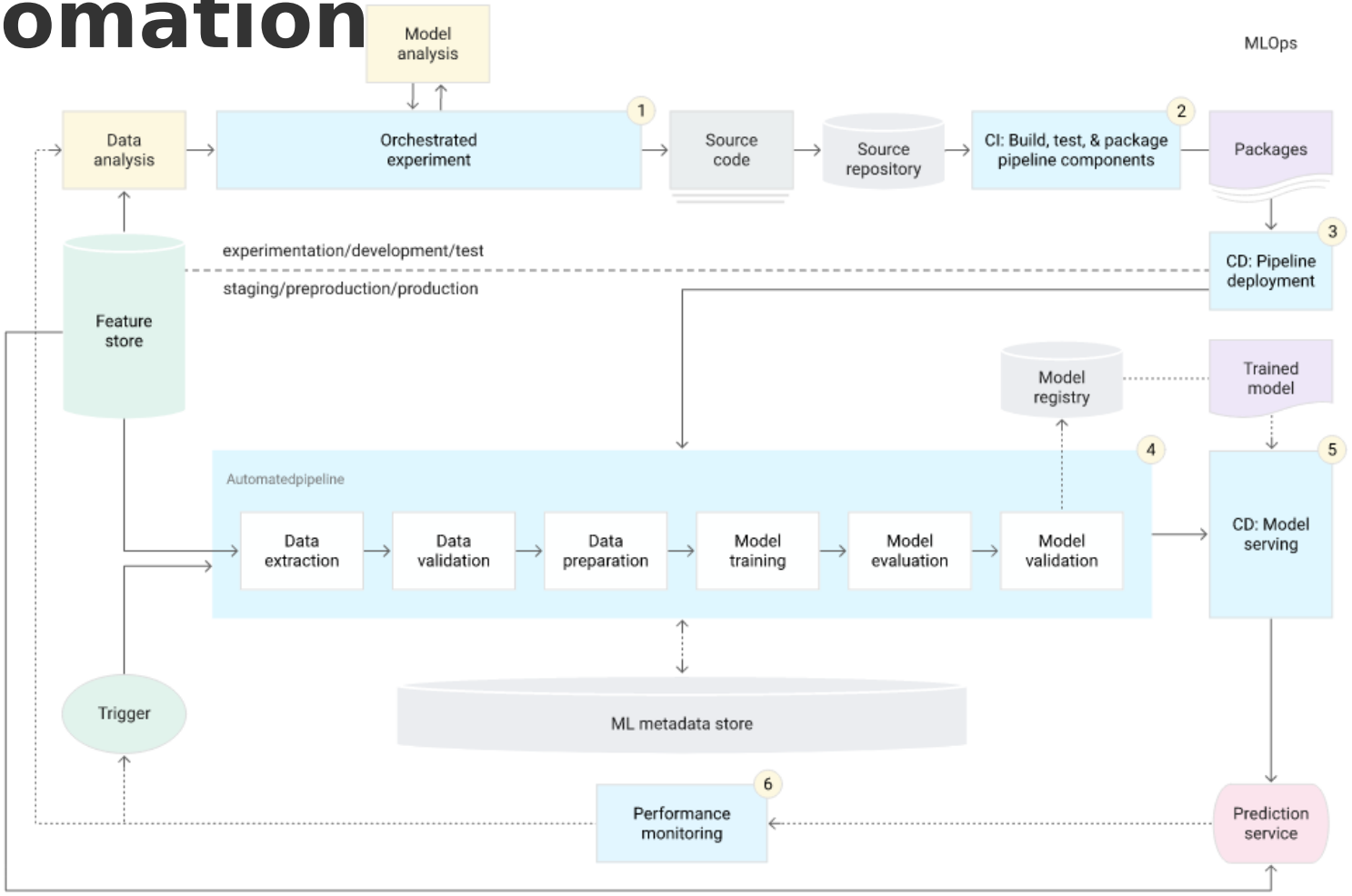


MLOps level 1: ML pipeline automation

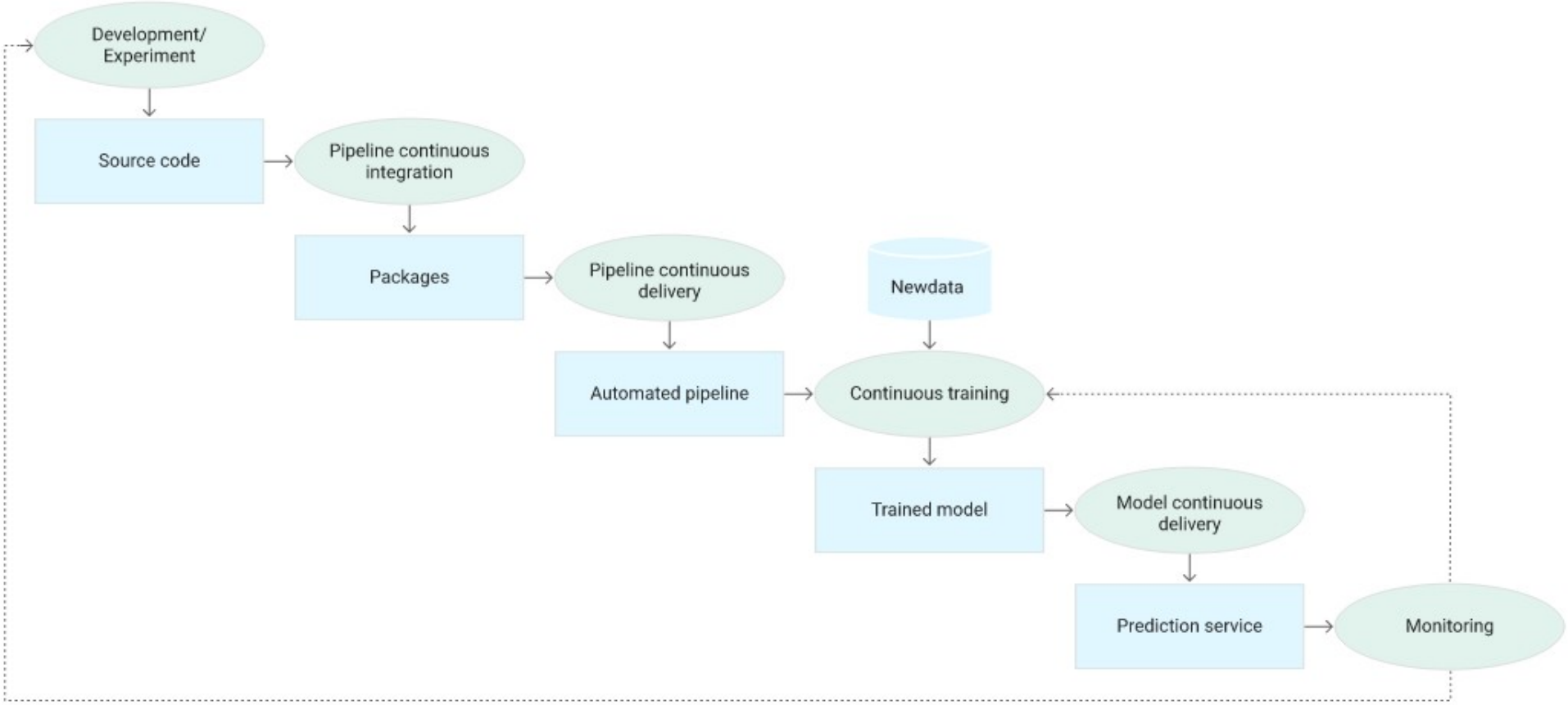
- Automated ML pipeline → rapid experiments
  - Same pipeline is reused in production
- Automatic (re)training in production + automatic deployment of new version
  - Data validation (before training) + Model validation (after)
  - Triggers: ad-hoc, scheduled, new data available, model drift...
- Training pipeline is modular, containerized
  - Reusable
  - Decoupled from runtime environment
- Deployment of whole training pipelines, not just trained models
- Feature store: centralized management of model inputs
- Metadata store: info about training pipeline executions (versions, artifacts, etc.)

→ **Quick update of existing models. New pipeline introduction may still be slow.**

MLOps level 2: CI/CD pipeline automation



MLOps level 2: CI/CD pipeline automation – CI/CD stages



MLOps level 2: CI/CD pipeline automation

- CI practices for pipeline development:
  - When new code is pushed…
  - Unit testing of feature engineering logic, data preparation methods etc.
  - Integration testing of pipeline components
  - Testing the trained model: convergence, extreme values etc.
- CD practices for pipeline deployment:
  - Automated deployment in test environment, manual trigger for deployment in production
  - Test input/output schema consistency of the new prediction service
  - Test model performance, prediction quality
- → **Full automation of both training pipeline and model deployment, testing.**

Tool landscape (not exhaustive)

- TFX (TensorFlow Extended): “end-to-end platform for deploying production ML pipelines”
- AirFlow: “platform to programmatically author, schedule and monitor workflows”
- Kubeflow: “Machine Learning Toolkit for Kubernetes”
- MLflow: “open source platform for the machine learning lifecycle”
- Seldon Core: “open source framework for easily and quickly deploying models and experiments at scale”
- *Commercial:* DataBricks managed MLflow, Amazon CodePipeline, Amazon SageMaker, Azure DevOps Pipelines, Azure ML, GCP Vertex AI, IBM Cloud Pak for Data

Final notes

- Note the gap between an ML PoC/prototype and a model in production
  - Most ML projects cannot bridge this gap!
- CI/CD style automation can help
  - Must take the ML specifics into account
- But, don’t overdo this!
  - If you are investing more into the ML platform development than ML use case development, then something is wrong… (or you are in the ML platform business ☺)